GROUP 4 PRESENTATION

Nguyen Duc Tri, Nghiem Xuan Khai, Vu Duc Manh, Nguyen Trung Hieu, Tran Long

PROOF OF STAKE

Is Proof of Stake the ultimate consensus mechanism for blockchain systems, or are new consensus models required to address its limitations?

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PROOF OF WORK: THE STARTING POINT

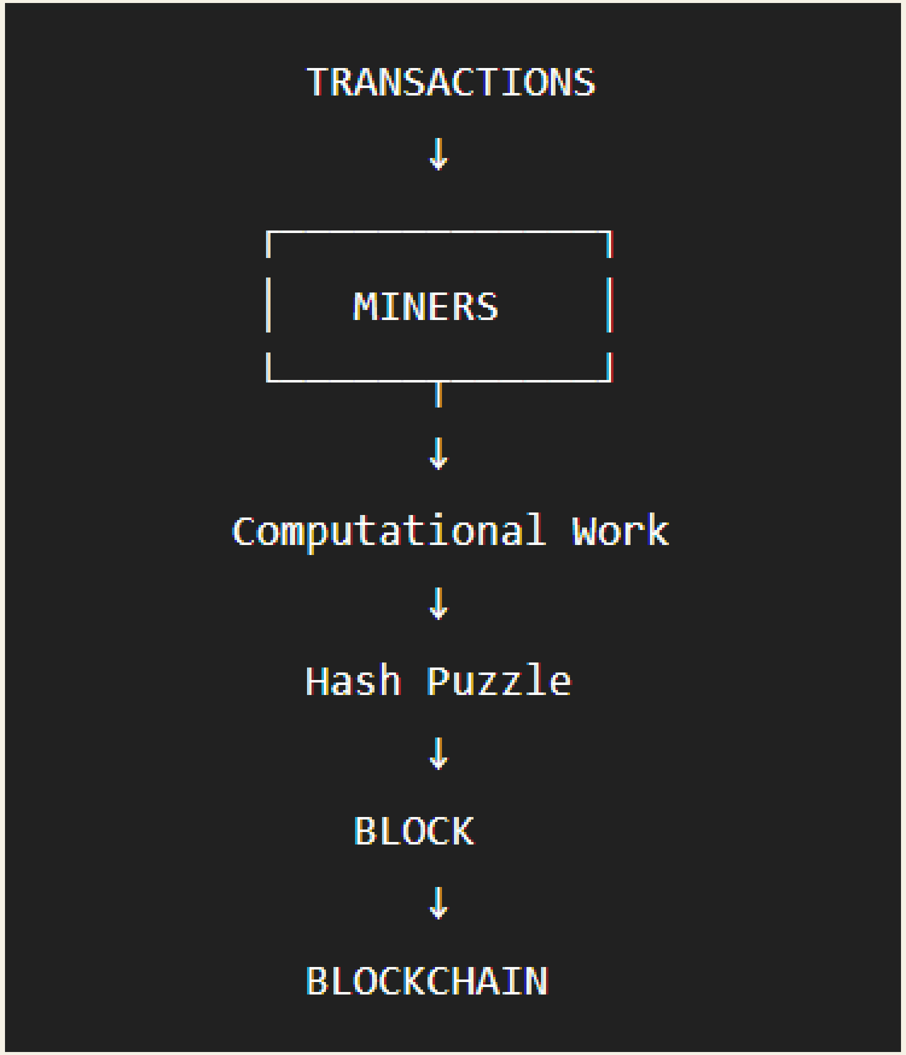
Core idea

Miners compete by performing computational work.

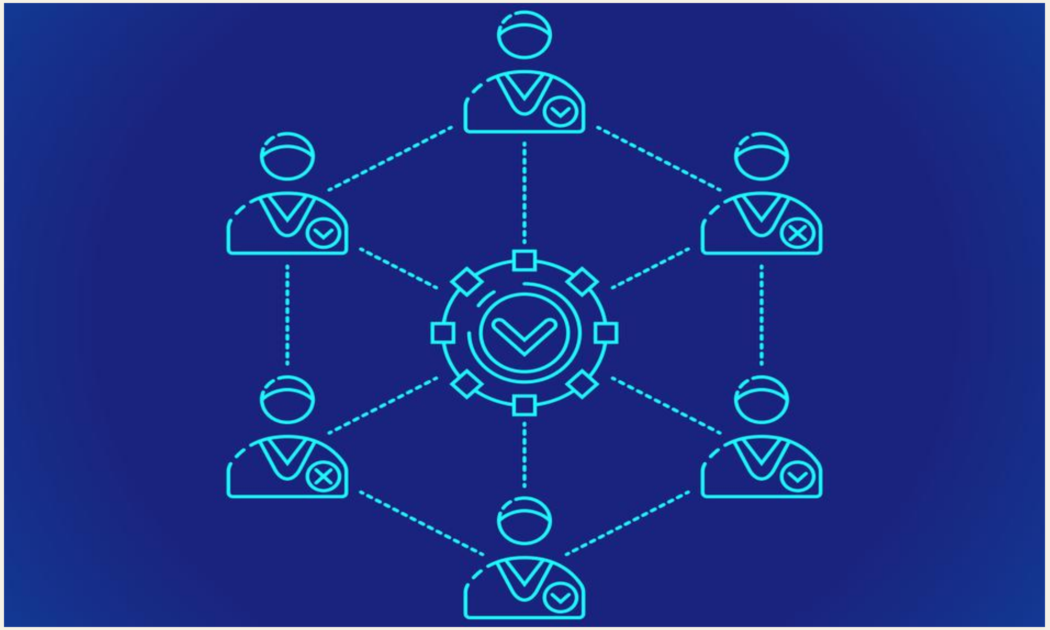
The valid block requires solving a cryptographic puzzle.

The accumulated work makes rewriting history costly.

⇒ PoW establishes consensus through computational cost.



WHY DOES CONSENSUS MATTER?



Consensus must provide

Agreement — nodes agree on the same state

Security — attackers cannot easily manipulate the ledger

Liveness — network continues to make progress

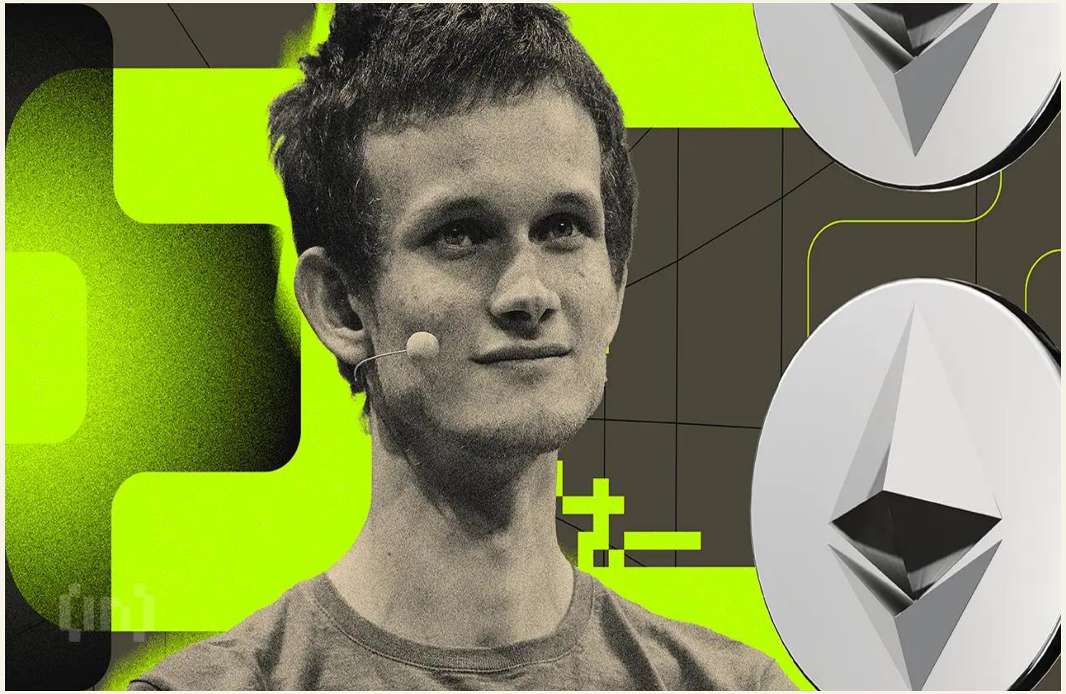
Blockchain

↓ Agreement

↓ Security

↓ Liveness

FROM POW TO POS



Proof of Work -> Proof of Stake
economic commitment

Security through computation

Computational work

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Hash puzzle

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Block

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Stake
  ↓
Validator
  ↓
Block
  ↓
Reward / Penalty
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WHAT IS POS?

Proof of Stake: Core Idea

Users

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Stake ETH / Tokens

↓

Become Validators

↓

Propose / Attest Blocks

↓

Consensus

↓

Reward

HOW DOES POS REACH CONSENSUS ?

Stake

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Validator Set

↓

Block Proposal

↓

Attestations

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Finality

1. Stake

Validator locks the asset.

2. Selection

Protocol selects validators to propose/attest.

3. Attestation

Other validators confirm the block.

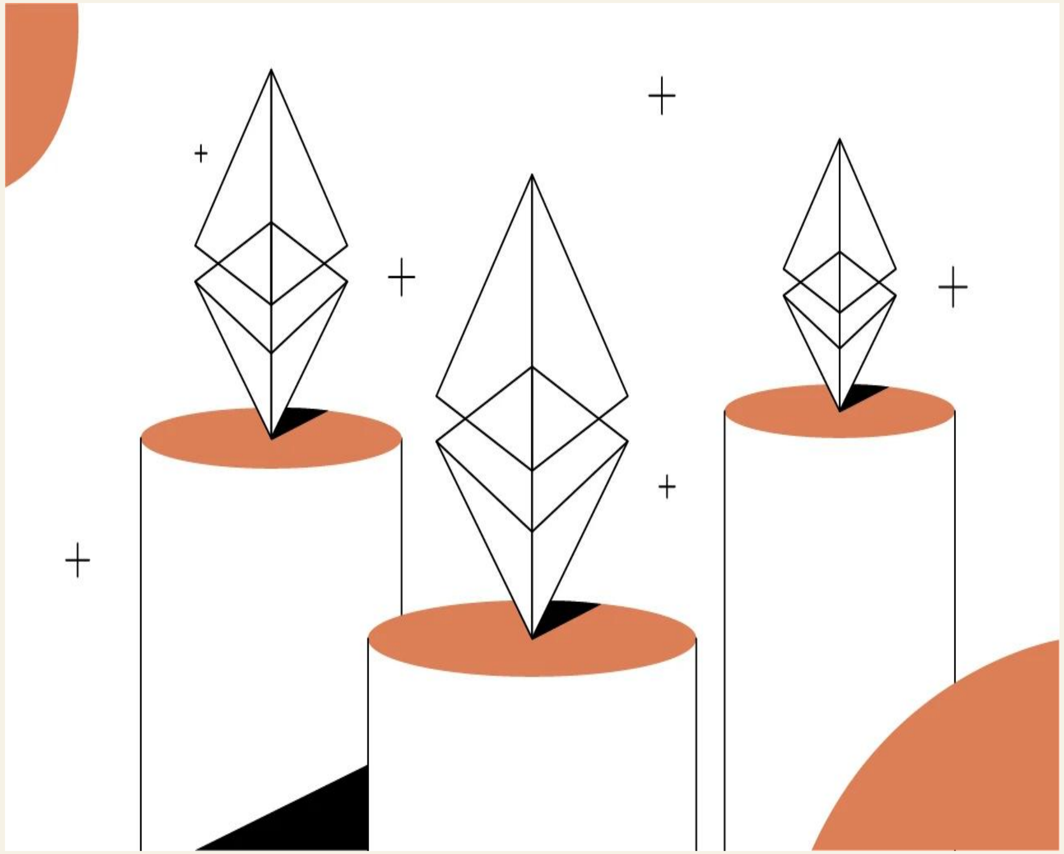
4. Finality

If all conditions are met, the block becomes finalized.

WHY POS IS POWERFUL?

PoS can provide strong security with dramatically lower energy requirements.

PoW	PoS
Energy-intensive	Energy-efficient
Mining hardware	Validator capital
Continuous computation	Economic security
Mining competition	Stake-based participation
Hardware centralization risk	Different centralization trade-offs



THE HIDDEN COST OF POS

PoS does not eliminate power concentration. It changes the source of power.

+ PoW: Capital => Hardware => Mining power

+ PoS:

Capital

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Stake

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Validator Power

THE SECURITY ASSUMPTIONS OF POS

QUESTION: Secure under what assumptions?

Economic assumptions

– Attackers incur significant economic costs.

Network assumptions

– Network must maintain communication/liveness.

Social assumptions

– In some extreme situations, social consensus may still be the last resort.

Operational assumptions

– Validators must operate the infrastructure stably.

ATTACK VECTORS / LIMITATIONS

Attack / Problem	Main Concern
51% / majority stake	Consensus manipulation
Validator concentration	Centralization
Long-range attacks	Historical chain manipulation
Nothing-at-stake	Conflicting histories
Stake concentration	Unequal influence
MEV	Economic centralization
Governance capture	Social/political attack

THE DEEPER PROBLEM

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                CONSENSUS
                    │
       ┌────────────┼────────────┐
       ↓            ↓            ↓
   Technical     Economic     Social
   Security      Incentives   Governance
       │            │            │
       └────────────┼────────────┘
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               TRUST SYSTEM
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Blockchain consensus is ultimately a mechanism for organizing trust under adversarial conditions.

BEYOND POS

Tendermint / CometBFT

BFT-oriented

+ Fast finality

+ Strong validator-based consensus

+ Suitable for application-specific chains

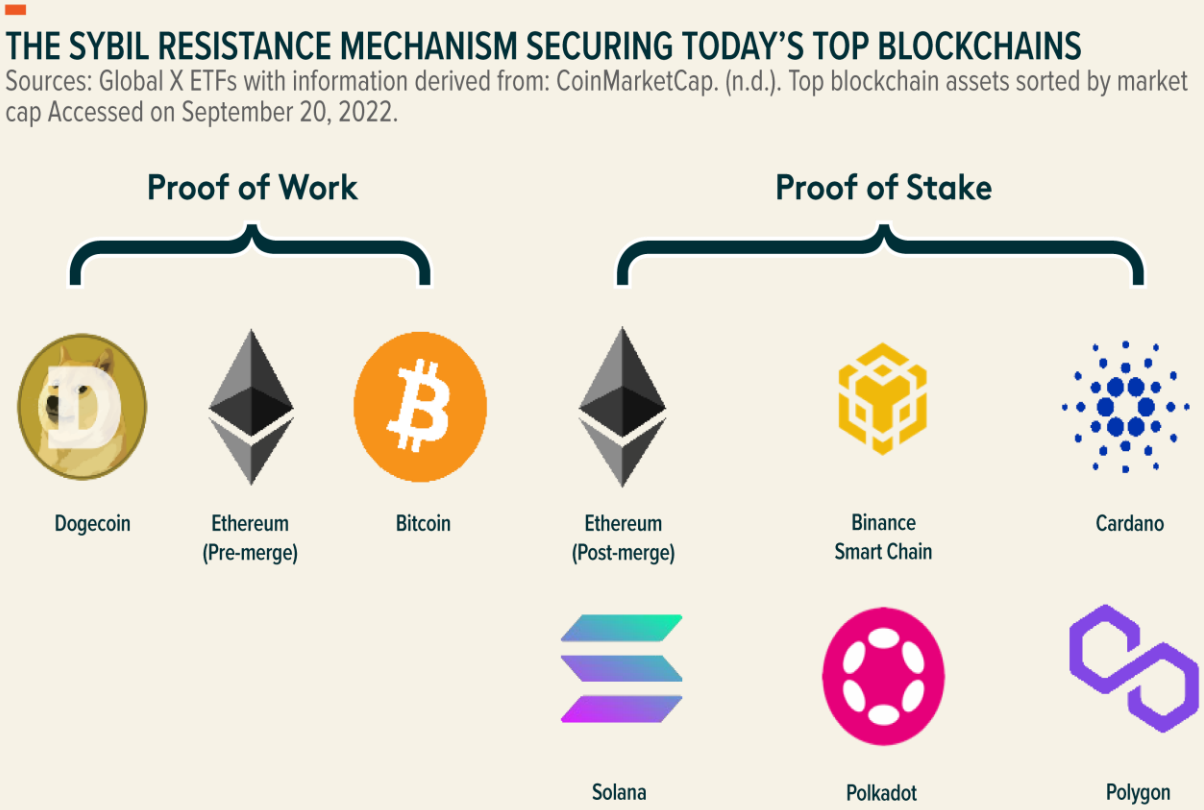
Avalanche

+ DAG / repeated randomized sampling

+ Probabilistic sampling

+ High scalability potential

+ Different security model from classical PoS/BFT



POS VS EMERGING MODELS

Property	PoW	PoS	Tendermint	Avalanche
Energy efficiency	❌	✅	✅	✅
Fast finality	Limited	✅	✅	✅
Capital requirement	Hardware	Stake	Stake	Stake
BFT orientation	Limited	Protocol-dependent	Strong	Different model
Scalability	Limited	High	High	High
Main trade-off	Energy	Stake concentration	Validator assumptions	Complexity/model assumptions

SO, IS POS THE ULTIMATE CONSENSUS ?

Not necessarily.

PoS solves important problems:

✓ Energy efficiency
✓ Economic security
✓ Decentralized validation
✓ Scalable consensus architecture

But introduces / retains challenges:

⚠️ Stake concentration
⚠️ Validator centralization
⚠️ Governance dependence
⚠️ Economic and social attack surfaces
⚠️ Application-specific trade-offs

CONCLUSION

Our Answer:

⇒ **Proof of Stake is not the ultimate consensus mechanism.**

Our prediction is:

Future blockchain systems will likely use different or hybrid consensus mechanisms depending on their security, scalability, economic, and governance requirements.

PoW

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PoS

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↓

BFT / DAG / Hybrid

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